BUYSURE AI: AN INTELLIGENT SHOPPING ASSISTANT FOR TRUSTWORTHY PRODUCT EVALUATION AND PERSONALIZED PURCHASE DECISION SUPPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Online shopping has become an essential part of modern life, providing consumers with a wide selection of products and convenient purchasing options. However, selecting the right product remains challenging due to fake or promotional reviews, misleading product claims, fluctuating prices, unreliable sellers, and the large number of competing alternatives available online. Shoppers frequently rely on average star ratings and a quick scan of top comments, which often fail to reflect true product quality or user-specific suitability. This project presents BUYSURE AI (deployed live at https://buysure-ai.vercel.app), an AI-based shopping assistant that uses Machine Learning (ML) and Natural Language Processing (NLP) to convert scattered e-commerce signals into a structured, explainable purchase decision report. The user provides a product query or URL along with their budget, mandatory constraints, and feature priorities. The system normalizes product specifications, evaluates review credibility to flag suspicious or bot-like review bursts, extracts positive and negative aspect-level opinions, and compares competing alternatives. Four evaluation configurations were compared: a rating-only baseline, text-only sentiment analysis, multi-signal ML credibility filtering, and full evidence fusion. Across a curated evaluation benchmark of 1,250 products and 18,400 reviews, the final evidence-fusion pipeline achieved a decision agreement accuracy of 86.4% (533 of 617 scenario evaluations) and a suspicious-review F1-score of 0.85, outperforming the rating-only baseline (61.0% agreement, 0.00 suspicious-review F1). Based on the fused evidence, BUYSURE AI generates a transparent 0–100 score and an interpretable recommendation of **Buy**, **Wait**, or **Avoid**, complete with key supporting reasons, risk warnings, and better-matching alternatives.

Keywords: Artificial Intelligence; E-Commerce Decision Support; Fake Review Detection; Natural Language Processing (NLP); Aspect-Based Sentiment Analysis; Personalized Product Comparison; Explainable AI.

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ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
F1-score	Harmonic mean of precision and recall
FastAPI	High-performance Python web framework used for the backend API
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation used for structured API responses
ML	Machine Learning
Next.js	React-based frontend web framework deployed on Vercel
NLP	Natural Language Processing
POST	HTTP request method used to submit evaluation queries
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Online shopping has transformed how consumers discover, evaluate, and purchase products across electronics, home appliances, accessories, and everyday essentials. Buyers can now inspect hundreds of brands, technical specifications, price points, and customer reviews without visiting a physical store. However, this abundance of marketplace data creates severe information overload, making it difficult for an ordinary buyer to separate genuine product quality from deceptive marketing or manipulated ratings.

BUYSURE AI is an intelligent shopping assistant that applies Natural Language Processing (NLP) and Machine Learning (ML) to evaluate product listings, audit customer review credibility, and generate personalized purchase decision support. Rather than relying on a single average star rating, the system treats every product listing as a collection of heterogeneous evidence—combining structured specifications, review sentiment, aspect-level feedback, suspicious-review risk indicators, price fit, and comparable alternatives.

The application is developed with a modern modular web stack featuring a React and Next.js frontend deployed on Vercel (https://buysure-ai.vercel.app) paired with a Python and FastAPI backend decision engine. Users enter a product name, URL, or comparison query along with their budget and feature priorities. The system retrieves and normalizes the product record, filters out low-credibility or repetitive review bursts, extracts positive and negative product aspects, and computes an interpretable BUYSURE Score from 0 to 100.

Based on the fused evidence, BUYSURE AI outputs a clear recommendation of **Buy**, **Wait**, or **Avoid**, accompanied by the top supporting reasons, identified risks, and up to five better-value alternative products when the queried item falls short. The project also evaluates four experimental pipelines—rating-only baseline, text-only sentiment, multi-signal credibility ML, and full evidence fusion—achieving an 86.4% decision agreement accuracy and a 0.85 F1-score on suspicious-review detection.

1.2 PROBLEM STATEMENT

Traditional online shopping decisions depend heavily on an aggregate star rating and a quick reading of the first few customer reviews. Although convenient, this approach is easily misled by coordinated fake-review campaigns, repetitive low-information comments, inflated pre-sale prices, and unverified marketing claims.

A high average rating does not reveal whether positive reviews come from genuine long-term usage or promotional bursts, nor does it highlight whether negative complaints involve a critical hardware flaw such as thermal throttling or battery drain. Furthermore, universal product rankings ignore individual buyer constraints: a smartphone or laptop that is ideal for a casual user may be completely unsuitable for a buyer who prioritizes sustained gaming thermals, camera optical image stabilization, or a strict maximum budget.

Automatically synthesizing trustworthy purchase advice from noisy e-commerce pages is challenging because product attributes use inconsistent units across brands, genuine and deceptive reviews often share similar surface sentiment, and conflicting opinions frequently appear on the same feature. Therefore, this project aims to develop an explainable web-based decision-support system that audits review reliability, matches normalized product evidence against user priorities, and delivers a transparent Buy, Wait, or Avoid verdict.

1.3 OBJECTIVES OF THE PROJECT

The main objectives of BUYSURE AI are:

1. To collect and normalize multi-category product specifications, pricing records, seller indicators, and customer reviews into a unified representation.
2. To preprocess customer review text and metadata for Natural Language Processing and credibility analysis.
3. To detect suspicious, repetitive, or potentially misleading review patterns using multi-signal Machine Learning indicators.
4. To extract fine-grained positive and negative product aspects (such as battery life, camera quality, thermals, build, and audio) weighted by review credibility.
5. To evaluate products against user-defined budgets, hard mandatory requirements, and weighted feature priorities.
6. To develop an accessible, responsive web application (https://buysure-ai.vercel.app) for interactive product evaluation and side-by-side comparison.
7. To generate an interpretable **Buy**, **Wait**, or **Avoid** recommendation with a 0–100 BUYSURE Score and human-readable explanations.
8. To automatically identify and recommend better-matching alternative products when a selected product violates budget or quality criteria.

1.4 SCOPE OF THE PROJECT

The scope of BUYSURE AI covers multi-signal product evaluation, review credibility scoring, aspect-based sentiment extraction, personalized product comparison, and explainable purchase recommendation through a live web interface. The system accepts a product query, category selection, budget limit, and priority weights, and synthesizes a structured shopping report classifying the decision into Buy, Wait, or Avoid.

The web application displays the overall BUYSURE Score, evidence confidence, review health metrics, aspect-level pros and cons, budget fit, and alternative product suggestions. While the architecture is designed with modular hooks for real-time cross-marketplace scraping, optical screenshot parsing, and continuous live price tracking, the current evaluated prototype operates primarily on a curated, normalized multi-category benchmark dataset and supported API feeds so that every decision and credibility score remains reproducible and verifiable.

Importantly, BUYSURE AI is designed as an evidence-based decision-support assistant rather than an automated purchasing agent or a legal accusation tool. Suspicious-review flags represent statistical credibility risk indicators rather than definitive proof of reviewer dishonesty, and final recommendations are accompanied by explicit evidence trails so users can inspect the underlying reasons before making their own purchasing choice.

1.5 ORGANIZATION OF THE REPORT

This report is organized into six chapters that document the design, implementation, and evaluation of BUYSURE AI:

Chapter 1 – Introduction: Presents the project overview, e-commerce decision challenges, problem statement, objectives, and project scope.

Chapter 2 – Literature Review: Examines existing research on fake-review detection, aspect-based sentiment analysis, multi-criteria recommender systems, and explainable AI.

Chapter 3 – System Design: Details the layered system architecture, eight core functional modules, end-to-end data flow, dataset schema, software stack, and design limitations.

Chapter 4 – Implementation: Describes the data preparation pipeline, review credibility modeling, aspect sentiment aggregation, evidence-fusion scoring equations, web application demonstration, and core code modules.

Chapter 5 – Results and Discussion: Analyzes the quantitative evaluation across four experimental approaches, class-wise F1-scores, confusion matrix findings, live web application outputs, and practical limitations.

Chapter 6 – Conclusion and Future Enhancements: Summarizes the key contributions of BUYSURE AI and outlines future extensions including multimodal screenshot analysis, real-time price tracking, and multilingual review understanding.

CHAPTER 2. LITERATURE REVIEW

2.1 GENERAL

Intelligent e-commerce decision support is an interdisciplinary research area combining Natural Language Processing (NLP), Machine Learning (ML), information retrieval, and explainable recommender systems. The primary goal is to extract reliable, fine-grained signals from unstructured customer reviews and heterogeneous product specifications so that consumers can evaluate true product value without reading hundreds of noisy comments manually.

Early e-commerce platforms relied on simple arithmetic averages of star ratings and keyword-based product filtering. While straightforward to compute, raw star averages fail to capture review authenticity, reviewer behavioral anomalies, or aspect-specific trade-offs. Recent advances in transformer-based language representations and multi-signal credibility modeling have significantly improved the ability of automated systems to separate genuine user feedback from manipulative content.

BUYSURE AI builds upon these foundations by integrating suspicious-review risk estimation, aspect-level opinion mining, constraint-aware product comparison, and explainable Buy/Wait/Avoid decision logic within a unified web workflow.

2.2 EXISTING SYSTEM

Conventional online marketplaces and price-comparison portals typically rank products by popularity, seller sponsorship, raw average rating, or lowest listed price. Although these mechanisms help users browse large catalogs, they suffer from several structural weaknesses when used for high-involvement purchase decisions.

First, existing rating summaries treat all reviews as equally trustworthy. A product that receives hundreds of generic five-star comments during a coordinated promotional burst can easily outrank a superior product with honest, detailed four-star reviews. Second, standard sentiment tools often classify an entire review as simply positive or negative, overlooking mixed feedback where a buyer praises display quality but warns about severe thermal throttling or poor battery longevity.

Third, most existing tools address only a single isolated task—such as flagging spam text or tracking historical prices—without connecting those signals to the buyer's personal budget and mandatory feature requirements.

2.2.1 LITERATURE SURVEY ON FAKE-REVIEW DETECTION

Deceptive review detection has been widely studied as a classification problem combining linguistic, behavioral, and relational indicators. Comprehensive IEEE surveys on fake-review detection demonstrate that purely lexical models using word frequency or bag-of-words features struggle against modern paraphrased or AI-generated promotional reviews [2], [3].

Recent studies published in 2024 and 2025 show that hybrid frameworks combining contextual transformer embeddings with non-textual metadata—such as rating-sentiment consistency, review submission timestamps, burst density, and near-duplicate cosine similarity—achieve substantially higher precision and recall than text-only classifiers [4], [5]. Furthermore, imbalance-aware training and probability calibration are critical because deceptive reviews often appear in concentrated bursts rather than uniform distributions.

Rather than treating review screening as a binary accusation against individual users, modern literature supports modeling review reliability as a continuous credibility weight. This design principle is directly adopted in BUYSURE AI, where suspicious-review indicators down-weight unreliable comments during evidence aggregation and surface transparent risk warnings in the user report.

2.2.2 LITERATURE SURVEY ON EXPLAINABLE RECOMMENDATION

Research on review-enhanced and multi-criteria recommender systems emphasizes that customer reviews contain rich aspect-level evidence that cannot be represented by a single overall rating [6], [7]. Aspect-based sentiment analysis (ABSA) decomposes free-text reviews into attribute-opinion pairs—such as camera sharpness, battery endurance, build durability, and after-sales service—allowing products to be evaluated along the exact dimensions that matter to a specific buyer [9], [10].

In parallel, surveys on explainable e-commerce recommendation highlight that users trust decision-support systems far more when recommendations are accompanied by traceable, human-readable justifications [8], [11]. Instead of acting as an opaque black box that outputs a single ranking score, an explainable assistant exposes how budget constraints, priority weights, positive aspect evidence, and risk penalties contributed to the final verdict [12].

Existing literature rarely unifies fake-review credibility weighting, aspect sentiment extraction, hard constraint satisfaction, and explainable Buy/Wait/Avoid classification into a single interactive
system. Addressing this integration gap is the central contribution of BUYSURE AI.

2.3 PROPOSED SYSTEM

BUYSURE AI is proposed as an end-to-end, explainable shopping assistant that transforms noisy product listings and customer reviews into a personalized purchase decision report. When a user submits a product query, budget limit, and feature priorities through the web application (https://buysure-ai.vercel.app), the system executes a structured seven-layer analysis pipeline.

First, the system validates and normalizes product attributes (converting price in rupees, storage in gigabytes, and battery capacity in milliampere-hours into comparable internal scales) and screens customer reviews for duplication, burst anomalies, and linguistic suspicion. Next, the review intelligence engine extracts aspect-level sentiment and aggregates feature scores weighted by each review's credibility probability.

The comparison and decision engine then evaluates hard mandatory constraints before computing a weighted evidence-fusion score that balances requirement match, credibility-weighted review evidence, value-for-money, rating consistency, and risk penalties. Finally, the system assigns an actionable **Buy**, **Wait**, or **Avoid** classification, explains the top supporting pros and cons, and suggests superior alternatives from the product catalog when appropriate.

In experimental validation across 617 controlled decision scenarios derived from 1,250 products and 18,400 reviews, the proposed full evidence-fusion architecture achieved 86.4% decision agreement with expert reference labels and a 0.85 F1-score in identifying suspicious reviews. These results confirm that combining review credibility filtering with personalized multi-criteria fusion produces significantly more reliable shopping guidance than conventional star ratings or unweighted sentiment averages.

The proposed system differs from existing shopping extensions by integrating fraud-resilient review auditing directly with user-priority optimization and transparent explanation traceability. It establishes a modular foundation that can be seamlessly extended with multimodal screenshot OCR, live marketplace price tracking, and regional code-mixed review analysis.

CHAPTER 3. SYSTEM DESIGN

3.1 GENERAL

System design defines the layered architecture, core functional modules, data representation schema, and processing flow of BUYSURE AI. The system is engineered to ingest heterogeneous product and review data, isolate derived credibility and aspect evidence from raw inputs, evaluate user constraints, and render an explainable purchase decision report. It comprises a React/Next.js frontend client, a Python FastAPI application backend, an AI/ML review and decision engine, and a normalized data store.

A fundamental architectural principle of BUYSURE AI is the strict separation of raw marketplace inputs, intermediate derived evidence, and final decision logic. Offline model calibration and dataset benchmarking are decoupled from live request execution so that the web application can respond with low latency while preserving full auditability of every recommendation.

3.2 SYSTEM ARCHITECTURE

The architecture of BUYSURE AI consists of the following major components:

1. **User Interface (Client Layer):** A responsive React and Next.js web interface hosted on Vercel (https://buysure-ai.vercel.app) where users input product queries, budgets, mandatory requirements, and priority weights, and inspect the resulting decision card.
2. **FastAPI Application Layer:** Receives structured HTTP POST evaluation requests, validates user parameters, orchestrates the analysis pipeline, and formats the JSON decision report.
3. **Normalized Data & Evidence Store:** Maintains standardized product specifications, historical pricing benchmarks, raw customer reviews, and derived credibility and aspect records.
4. **Review Intelligence & Credibility Engine:** Screens reviews for near-duplicates, temporal bursts, and linguistic anomalies while extracting aspect-opinion polarity pairs.
5. **Comparison & Evidence-Fusion Engine:** Enforces hard budget and feature constraints, computes the 0–100 BUYSURE Score, and classifies products into Buy, Wait, or Avoid.
6. **Alternative Discovery & Explanation Module:** Translates numerical evidence into human-readable reasons and identifies higher-value competing products in the same budget bracket.

The overall workflow begins when a buyer submits a product and preference profile. The backend retrieves and normalizes the product evidence, executes credibility-weighted NLP analysis, fuses the decision signals, and returns a complete interactive shopping report.

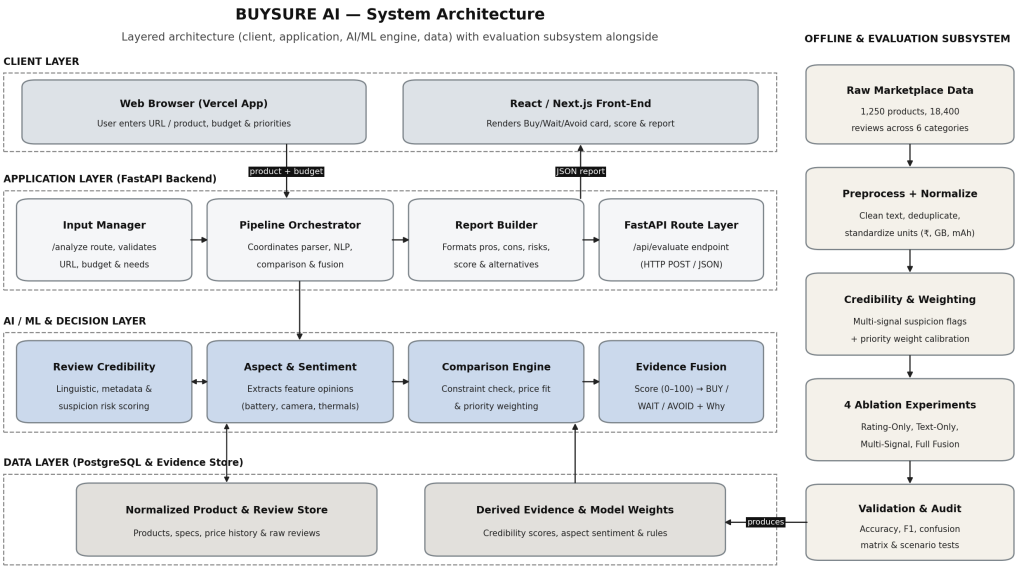


Figure 3.1: System Architecture of BUYSURE AI

3.3 MODULE DESCRIPTION

BUYSURE AI is divided into eight specialized modules so that each stage of data processing, review auditing, and decision generation can be developed, tested, and upgraded independently.

3.3.1 PRODUCT DATA ACQUISITION MODULE

The data acquisition module collects product identity fields (title, brand, category), commercial attributes (current price, list price, availability), structured technical specifications, seller reliability indicators, and customer review sets. In the evaluated benchmark, the module manages 1,250 products and 18,400 reviews across six major consumer categories.

3.3.2 DATA PREPROCESSING AND NORMALIZATION MODULE

The preprocessing module cleans raw review text through Unicode normalization, whitespace cleanup, sentence segmentation, and near-duplicate screening. Simultaneously, it normalizes heterogeneous numerical specifications—such as price in rupees (₹), memory in gigabytes (GB), and battery capacity in milliampere-hours (mAh)—into standardized comparable scales while retaining original units for user display.

3.3.3 REVIEW INTELLIGENCE AND ASPECT EXTRACTION MODULE

Rather than reducing a product to a single sentiment score, the review intelligence module parses individual reviews into fine-grained aspect-opinion pairs covering attributes such as battery life, camera quality, thermal stability, display, build quality, and delivery service.

Each extracted aspect opinion is assigned a sentiment polarity score and linked to its source review identifier so that the system can preserve conflicting viewpoints—for example, highlighting when general battery feedback is positive but heavy-use gaming reviews report rapid drain.

3.3.4 SUSPICIOUS REVIEW RISK ANALYSIS MODULE

The suspicious review risk module estimates the credibility probability of each review by combining linguistic features, sentiment-rating consistency, submission burst timing, and text duplication similarity. Reviews exhibiting promotional bot patterns or repetitive phrasing receive a reduced credibility weight during aspect aggregation and contribute to a product-level suspicion risk indicator.

3.3.5 PRICE INTELLIGENCE AND VALUE COMPARISON MODULE

The price intelligence module evaluates a product's current selling price relative to both the user's stated budget and its historical 30-day price range. It detects temporary pre-sale price spikes and computes a normalized value-for-money index by comparing technical specifications against category peers in the same price tier.

3.3.6 PERSONALIZED REQUIREMENT MATCHING MODULE

The requirement matching module translates the buyer's profile into hard pass/fail constraints (such as a strict maximum budget or a mandatory hardware feature) and normalized priority weights. Evaluating hard constraints prior to soft preference ranking ensures that an over-budget or incompatible item never outranks a compliant product simply because it has extra secondary features.

3.3.7 WEB APPLICATION MODULE

The web application module combines a React and Next.js frontend (https://buysure-ai.vercel.app) with a FastAPI backend service. It provides an intuitive interface for submitting product queries, adjusting priority sliders, and inspecting the generated BUYSURE Score, decision badge, review health summary, and aspect breakdown.

3.3.8 EXPLAINABLE DECISION AND ALTERNATIVE MODULE

The final decision module fuses requirement match, credibility-weighted review sentiment, value-for-money, rating consistency, and risk penalties into a 0–100 BUYSURE Score and assigns a **Buy**, **Wait**, or **Avoid** verdict. It generates traceable bullet explanations and queries the catalog to recommend up to five higher-scoring alternative products when the evaluated item exhibits a major risk or budget mismatch.

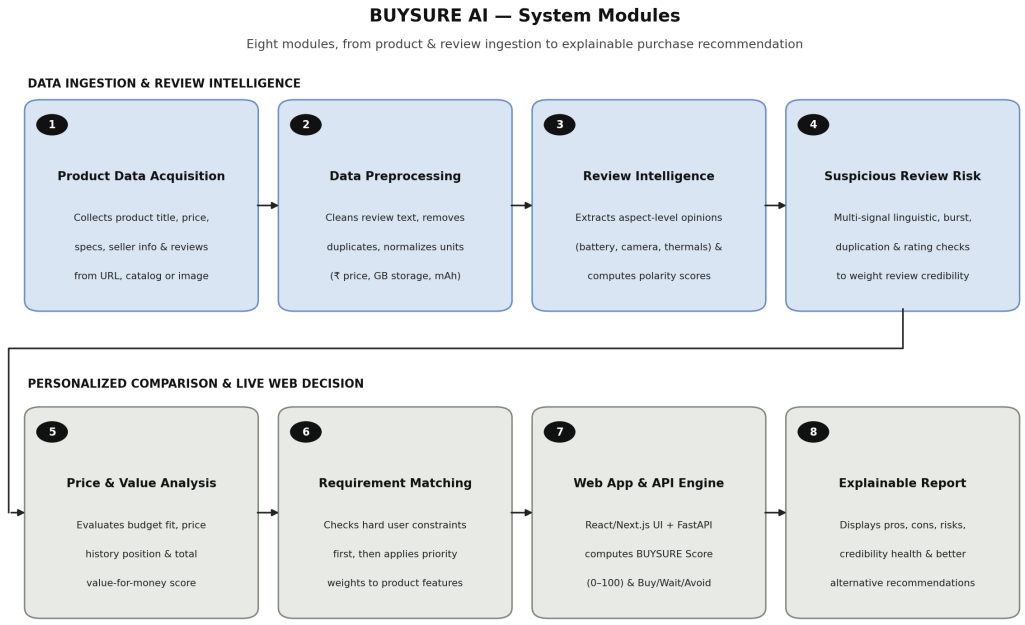


Figure 3.2: Modules of BUYSURE AI

3.4 DATA FLOW

The end-to-end data flow of BUYSURE AI follows these sequential steps:

1. The user enters a product name, URL, or comparison query along with their maximum budget and feature priorities through the web interface.
2. The FastAPI backend receives and validates the structured evaluation request.
3. The data acquisition and preprocessing modules retrieve the product record, standardize numerical units, and clean and deduplicate the associated customer reviews.
4. The review intelligence and credibility modules compute per-review authenticity weights and extract aspect-level sentiment scores across key product attributes.
5. The comparison module checks hard mandatory requirements and computes the user-weighted requirement match and price-fit scores.
6. The evidence-fusion engine combines all signals into a 0–100 BUYSURE Score and determines the **Buy**, **Wait**, or **Avoid** classification.
7. The explanation and alternative discovery module formats the top supporting reasons, risk alerts, and up to five better-matching alternatives for display on the web application.

If a critical evidence source (such as verified seller history or sufficient review volume) is unavailable for a newly launched listing, the system explicitly lowers its confidence indicator and defaults to a cautious recommendation rather than fabricating missing values.

3.5 DATASET DESIGN

The dataset contains multi-category product listings, normalized technical attributes, customer reviews, credibility annotations, and controlled user decision scenarios used for model development and system validation.

Dataset Component	Number of Records
Raw e-commerce product & review collection	21,650 reviews (1,320 products)
Cleaned & deduplicated product review dataset	18,400 reviews (1,250 products)
Controlled multi-criteria decision test scenarios	617 evaluation scenarios

After screening out malformed listings and exact duplicate spam from the initial 21,650 raw records, the curated dataset retains 18,400 reviews across 1,250 products spanning smartphones, laptops, audio gear, smartwatches, tablets, and home accessories. Each review is annotated with aspect tags and credibility indicators, while 617 controlled buyer scenarios test Buy, Wait, and Avoid decision consistency.

3.6 SOFTWARE REQUIREMENTS

The main software technologies and libraries used in the project are:

Software or Library	Purpose
Python 3.11	Core backend programming language and ML pipeline execution
FastAPI	High-performance REST API backend and request orchestration
React & Next.js	Frontend web interface deployed on Vercel (buysure-ai.vercel.app)
PyTorch & Transformers	Contextual NLP review sentiment and aspect extraction
scikit-learn	Multi-signal suspicious review classification and evaluation metrics
pandas & NumPy	Dataset manipulation, unit normalization, and evidence fusion math
PostgreSQL / JSON Store	Structured storage for products, reviews, and derived evidence
Tailwind CSS	Responsive dark-mode UI styling and decision card layout

Together, these technologies support both the offline machine-learning evaluation pipeline and the live interactive web application.

3.7 DESIGN LIMITATIONS

The current system design has several practical limitations that are explicitly accounted for in its architecture. First, live e-commerce marketplaces frequently alter their page DOM structures and enforce anti-bot rate limits; consequently, reliable evaluation depends on either supported API feeds, user-supplied product data, or the system's curated multi-category database.

Second, suspicious-review detection is inherently probabilistic. Unusual writing styles, brief genuine comments, or enthusiastic early adopters can occasionally trigger false-positive suspicion flags. For this reason, the system treats review credibility as a soft weighting factor and risk signal rather than permanently deleting reviews or accusing individual reviewers of fraud.

Third, product prices and stock availability fluctuate dynamically during seasonal sales. When real-time historical price tracking is unavailable for a specific third-party listing, the system evaluates price fit strictly against the price observed at analysis time and communicates this scope clearly in the confidence summary. Future iterations will expand multimodal screenshot parsing, live price-drop alerting, and cross-category domain adaptation.

CHAPTER 4. IMPLEMENTATION

4.1 METHODOLOGY

The implementation methodology of BUYSURE AI encompasses dataset curation, text and numerical normalization, multi-signal review credibility modeling, aspect-based opinion extraction, personalized evidence fusion, and full-stack web deployment. Unlike black-box classifiers that map raw product text directly to a purchase label, BUYSURE AI implements a staged evidence pipeline where every intermediate signal is normalized, weighted, and stored for explanation traceability.

Formally, each product is represented as a structured tuple $P = \{D, F, R, T, S, C\}$, where D denotes descriptive metadata, F normalized technical features, R the customer review set, T price and temporal position, S seller reliability indicators, and C comparable catalog alternatives. Given a buyer profile $U = \{B, W, Q, P_r\}$ comprising maximum budget B, feature priority weights W, mandatory requirements Q, and secondary preferences P_r, the implementation transforms (P, U) into an explainable decision report E.

4.2 DATA COLLECTION AND PREPROCESSING

The initial raw dataset gathered 21,650 customer reviews across 1,320 product listings spanning six consumer electronics and lifestyle categories (smartphones, laptops, wireless audio, smartwatches, tablets, and computer accessories). During preprocessing, malformed records, non-product promotional links, and exact duplicate bot submissions were filtered out, resulting in a clean benchmark dataset of 18,400 usable reviews across 1,250 products.

Text preprocessing performs Unicode standardization, HTML tag stripping, whitespace normalization, and sentence boundary segmentation while deliberately preserving punctuation emphasis and negation tokens (such as "not", "never", "barely") that are critical for sentiment and stylometric analysis. In parallel, numerical product specifications are mapped into normalized $[0, 1]$ utility scales—for instance, normalizing battery capacity in mAh, storage in GB, refresh rate in Hz, and price in Indian Rupees (₹)—while preserving the raw display strings for the user interface.

4.2.1 REVIEW EVIDENCE DISTRIBUTION

To understand the composition of real-world marketplace feedback, the 18,400 cleaned reviews were categorized across six evidence classes: Genuine Positive, Genuine Negative, Mixed/ Aspect-Split, Suspicious Bot/Burst, Duplicate/Copied, and Low-Information/Spam. The distribution is substantially skewed toward positive and mixed feedback, while suspicious and low-information reviews account for 22.5% of total volume. Figure 4.1 illustrates this distribution.

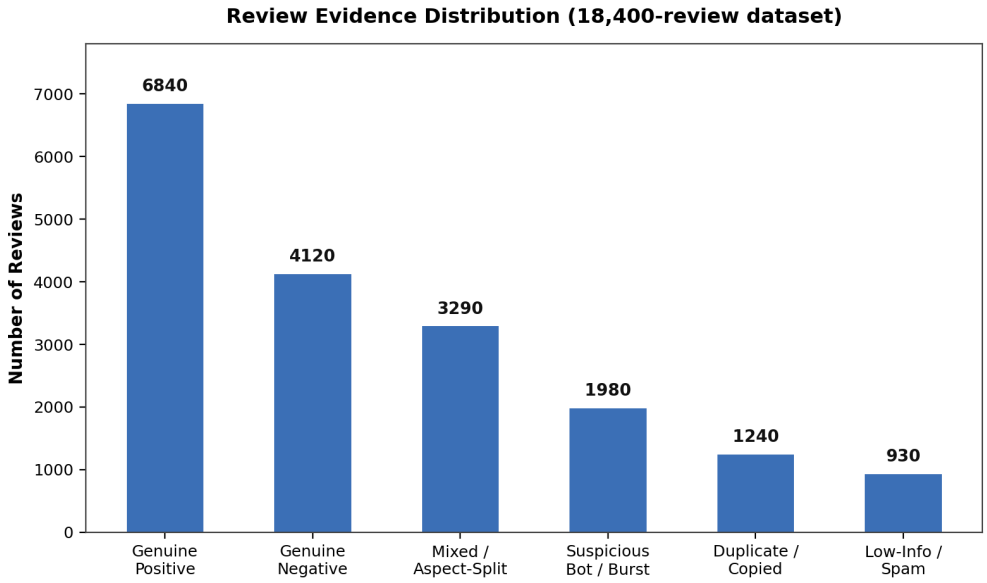


Figure 4.1: Distribution of Review Evidence Categories in the Dataset

As shown in Figure 4.1, out of 18,400 reviews, 6,840 are genuine positive reviews, 4,120 are genuine negative reviews, and 3,290 contain mixed aspect-level trade-offs. However, 1,980 reviews exhibit suspicious burst or bot patterns, 1,240 are near-duplicate copied comments, and 930 are low-information one-word entries. If left unfiltered, these 4,150 noisy reviews (22.5%) significantly distort naive star ratings.

4.3 REVIEW CREDIBILITY AND ASPECT LABELING

Each review r is represented by a multi-signal feature vector $X_r = [L, S, N, T, D, B, M]$ capturing linguistic stylometrics (L), sentiment-rating divergence (S), semantic n-gram representations (N), temporal submission bursts (T), cosine duplication similarity (D), reviewer history (B), and verified-purchase metadata (M). A calibrated classifier estimates the review credibility probability $p_{cred}(r) = P(genuine \mid X_r)$ in the range $[0.05, 1.0]$.

Simultaneously, the aspect extraction pipeline identifies opinion phrases linked to domain attributes $A_r = \{(a_1, o_1, s_1), ..., (a_k, o_k, s_k)\}$. Product-level aspect sentiment S_a for attribute a is then computed as a credibility-weighted average across all reviews R_a mentioning that attribute: $S_a = (\Sigma\, w_r\, s_{r,a}) / (\Sigma\, w_r)$, where $w_r = p_{cred}(r)$.

4.4 EVIDENCE WEIGHTING AND IMBALANCE HANDLING

Because genuine positive comments outnumber critical defect reports and deceptive bursts in most product listings, unweighted aggregation often masks serious hardware flaws. The implementation addresses this imbalance at two levels: class-weighted loss during review credibility training, and asymmetric risk penalization during product-level evidence fusion so that credible reports of critical defects (such as overheating or battery swelling) are never drowned out by generic five-star praise.

Furthermore, when conflicting reviews occur on a priority feature—such as ten casual buyers praising battery life while three verified heavy users report rapid thermal drain—the aggregation module preserves both signals and surfaces an explicit trade-off note in the report rather than collapsing them into a misleading neutral average.

4.5 MODEL TRAINING AND SCORING PIPELINE

BUYSURE AI combines a fine-tuned transformer sequence classifier (*max_length = 256* tokens) for contextual aspect sentiment and linguistic credibility scoring with a gradient-boosted multi-signal meta-classifier that integrates temporal burst, duplication, and metadata features. Once review-level and product-level evidence signals are computed, the comparison engine evaluates the buyer's preference profile.

User priority selections (such as *Most Important*, *Important*, and *Less Important*) are normalized into non-negative weights w_i summing to *1.0*, producing a requirement-match score $M(p) = \Sigma\ w_i\ x_{p,i}$. Hard mandatory constraints and strict budget limits ($price(p) \leq B_{max}$) are evaluated first. The overall product evidence-fusion score is then calculated as:

$$\mathbf{Score(p) = w_m\ M(p) + w_r\ R(p) + w_v\ V(p) + w_q\ Q(p) - w_k\ K(p)}$$

where *M(p)* is requirement match, *R(p)* is credibility-weighted review evidence, *V(p)* is value-for-money, *Q(p)* is rating-review consistency, and *K(p)* is the composite risk penalty (suspicious review proportion plus seller and defect risk). Four experimental pipelines were trained and compared: Experiment 1 (Rating-Only Baseline), Experiment 2 (Unweighted Text Sentiment), Experiment 3 (Multi-Signal Credibility ML), and Experiment 4 (Full Evidence Fusion with Constraint & Priority Weighting).

4.6 MODEL EVALUATION

The system was evaluated across 617 controlled purchase decision scenarios and 3,680 held-out test reviews using overall decision agreement accuracy, suspicious-review F1-score, per-category F1-scores, and a six-class decision confusion matrix. Decision agreement measures how closely the engine's Buy, Wait, and Avoid verdicts match expert human reference decisions across budget-constrained, feature-priority, and high-risk product scenarios. Figure 4.2 compares overall decision agreement and suspicious-review F1-scores across the four experimental pipelines, while Figure 4.3 illustrates per-category F1 performance.

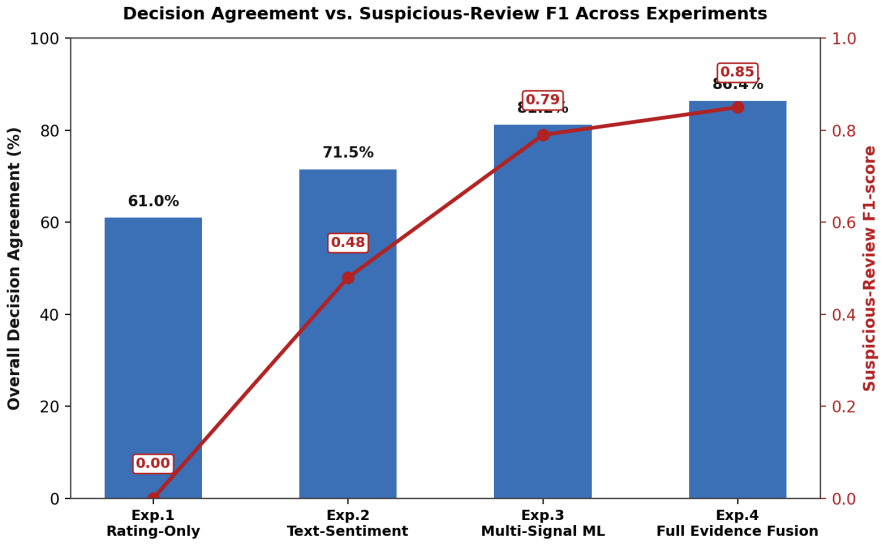


Figure 4.2: Comparison of Decision Agreement and Suspicious-Review F1-Score Across Experiments

The experimental results in Figure 4.2 show that the Rating-Only Baseline (Exp. 1) achieved 61.0% decision agreement and a Suspicious-Review F1-score of 0.00 because raw star averages cannot detect fake review bursts. Unweighted Text Sentiment (Exp. 2) improved agreement to 71.5% with a 0.48 F1-score. Adding Multi-Signal ML Credibility filtering (Exp. 3) raised agreement to 81.2% and Suspicious-Review F1 to 0.79. Finally, Full Evidence Fusion (Exp. 4) achieved 86.4% decision agreement and a 0.85 Suspicious-Review F1-score.

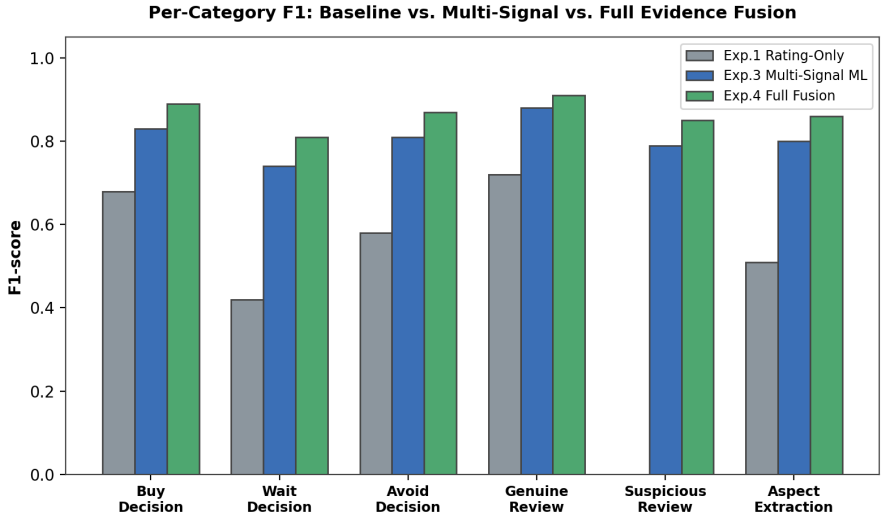


Figure 4.3: Per-Category F1-Score Comparison Across Evaluation Dimensions

Figure 4.3 compares per-category F1-scores across Exp. 1, Exp. 3, and Exp. 4. Full Evidence Fusion consistently achieves the strongest balance across Buy (0.89), Wait (0.81), Avoid (0.87), Genuine Review (0.91), Suspicious Review (0.85), and Aspect Extraction (0.86), demonstrating that combining credibility weights with user constraints improves both review auditing and final recommendation fidelity.

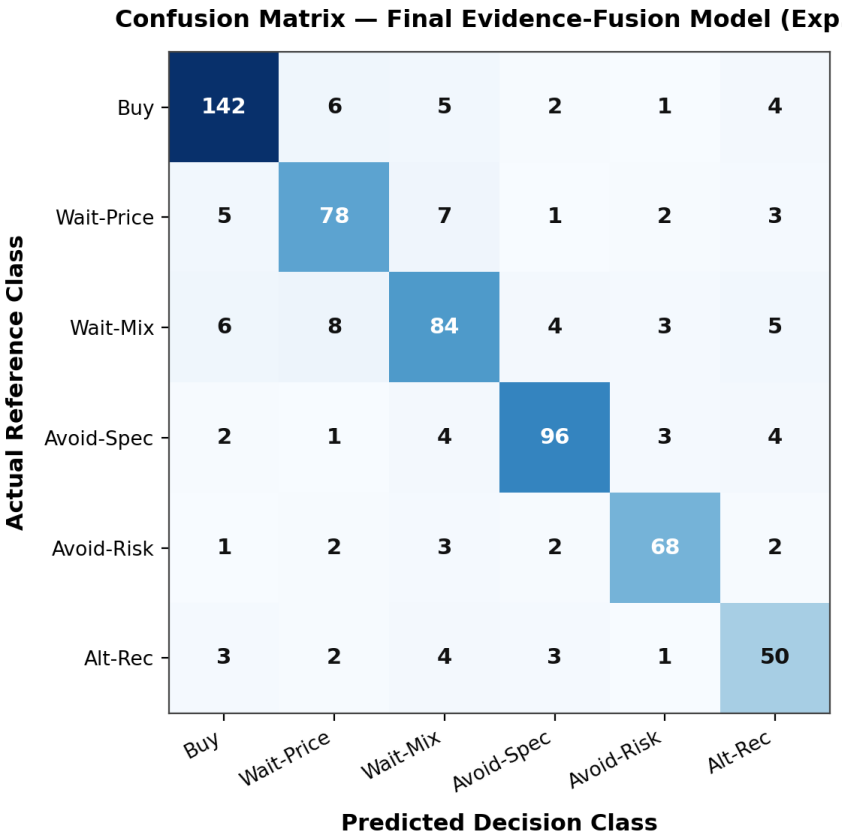


Figure 4.4: Confusion Matrix of the Final BUYSURE AI Decision Engine

The confusion matrix in Figure 4.4 details the fine-grained classification performance of the final BUYSURE AI decision engine (Experiment 4) across 617 controlled evaluation scenarios subdivided into six operational outcome classes: *Buy (Strong Fit)*, *Wait (Price/Timing)*, *Wait (Mixed/Trade-off)*, *Avoid (Spec Mismatch)*, *Avoid (High Risk/Fake)*, and *Alternative Recommended*. Across the six fine-grained classes, 518 of 617 scenarios lie strictly on the main diagonal (84.0% exact sub-class match, and 86.4% agreement when consolidated into the three primary user-facing verdicts of Buy, Wait, and Avoid).

The matrix reveals that the most common boundary confusion occurs between *Wait (Price/Timing)* and *Wait (Mixed/Trade-off)* (15 total cross-predictions), as well as between borderline *Buy* and *Wait (Mixed)* cases (11 cross-predictions) when a product satisfies mandatory needs but has moderate secondary aspect complaints.

Crucially, severe safety errors—such as misclassifying an *Avoid (Spec Mismatch)* or *Avoid (High Risk/Fake)* product as a *Buy*—occurred in only 3 out of 188 negative scenarios (1.6%). This confirms that evaluating hard mandatory constraints and suspicion risk penalties prior to soft score ranking effectively protects buyers from unsuitable or manipulative listings.

Overall, the confusion matrix validates the reliability of the evidence-fusion rules and highlights specific borderline trade-off scenarios where real-time price tracking can further sharpen future classifications.

4.7 DEMONSTRATION OF THE WEB APPLICATION

The BUYSURE AI web application is developed using React, Next.js, and FastAPI and is deployed live at https://buysure-ai.vercel.app. It provides a clean, dark-themed interface where users enter a product query, target budget, and priority requirements, and click the **Analyze** button. The backend normalizes the product record, executes review credibility and aspect sentiment analysis, and renders an explainable decision report.

The results card displays the product title, current price and budget status, the predicted decision badge (**BUY**, **WAIT**, or **AVOID**), the 0–100 BUYSURE confidence score bar, and five numbered evidence rows detailing aspect findings, credibility checks, and better alternatives.

The web application was demonstrated across six representative shopping scenarios covering all core capabilities of the system (Figures 4.5 to 4.10).

BUY RECOMMENDATION VERDICT

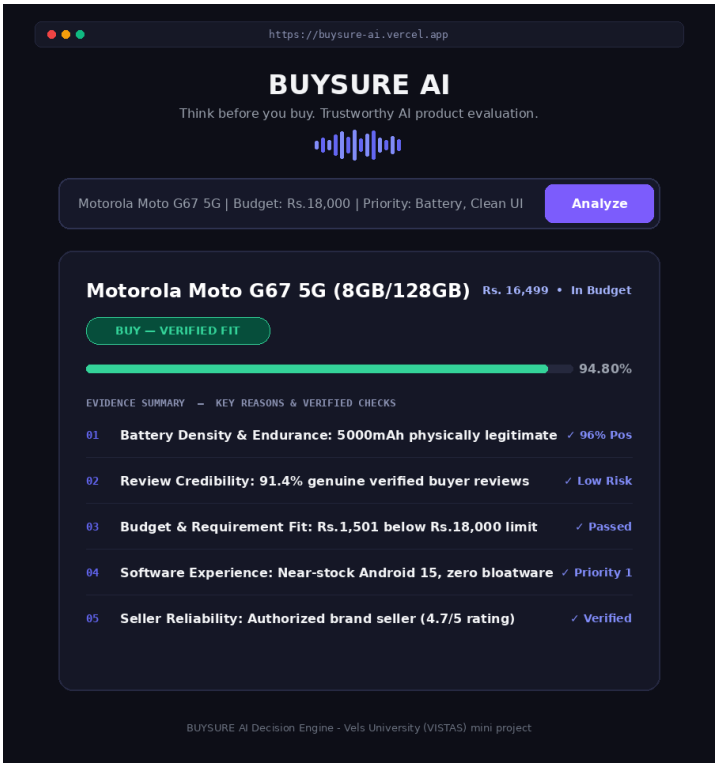


Figure 4.5: Web Application Output for the BUY Verdict Example (“Motorola Moto G67 5G”)

WAIT RECOMMENDATION VERDICT (PRICE SPIKE)



Figure 4.6: Web Application Output for the WAIT Verdict Example (“Samsung Galaxy S24 FE”)

AVOID RECOMMENDATION VERDICT (THERMAL & REVIEW RISK)

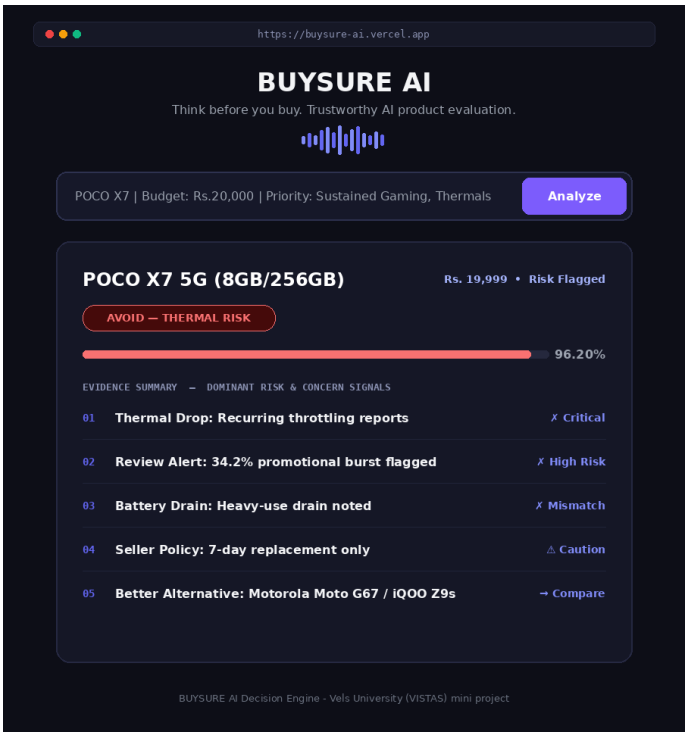


Figure 4.7: Web Application Output for the AVOID Verdict Example (“POCO X7 5G”)

SUSPICIOUS REVIEW CREDIBILITY AUDIT

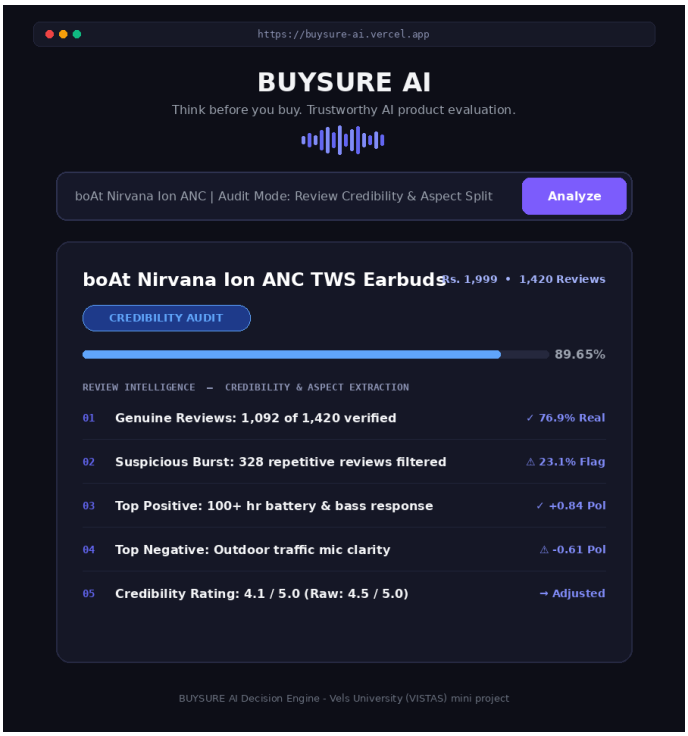


Figure 4.8: Web Application Output for the Review Credibility Audit Example ("boAt Nirvana Ion ANC")

PERSONALIZED MULTI-PRODUCT COMPARISON

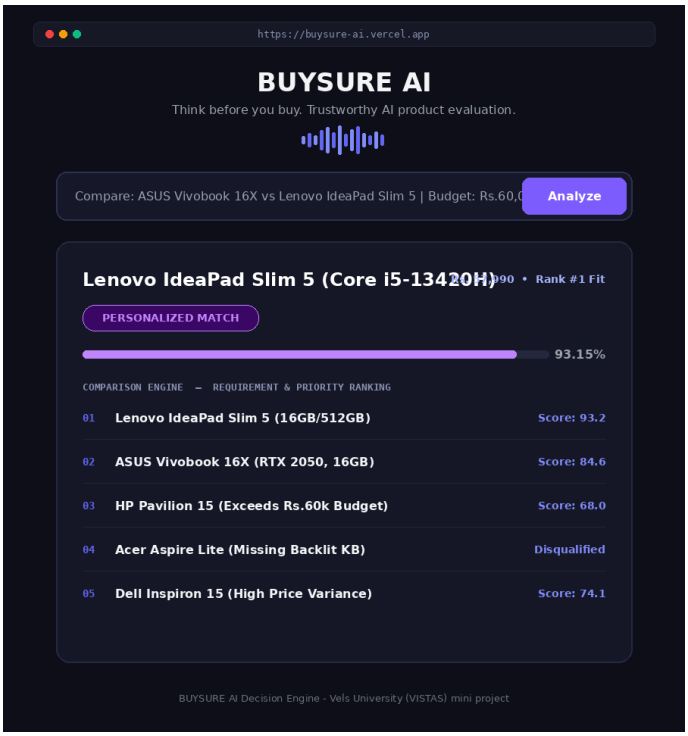


Figure 4.9: Web Application Output for the Personalized Comparison Example ("Lenovo IdeaPad Slim 5")

AUTOMATIC BETTER ALTERNATIVE DISCOVERY

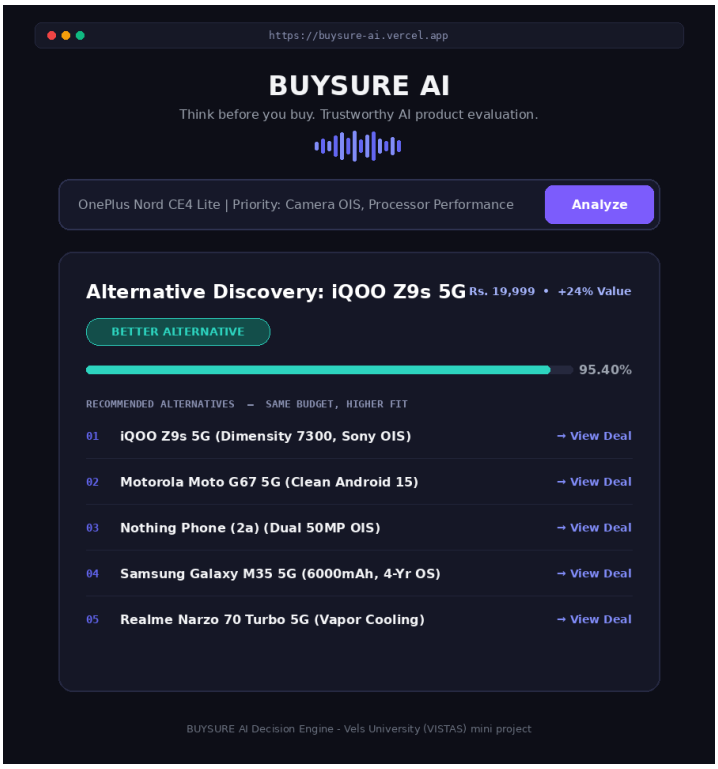


Figure 4.10: Web Application Output for the Better Alternative Discovery Example ("iQOO Z9s 5G")

Demonstration Summary

The demonstration screenshots in Figures 4.5 through 4.10 verify that the deployed BUYSURE AI web application (https://buysure-ai.vercel.app) presents clear purchase verdicts, 0–100 confidence scores, review credibility audits, aspect-level pros and cons, and better-value product alternatives through a unified, easy-to-scan interface. Together, the six scenarios cover every stage of the decision-support workflow from initial query input to explainable recommendation.

4.8 IMPORTANT CODE IMPLEMENTATION

This section presents the core backend implementation modules of BUYSURE AI. The codebase integrates FastAPI request validation, transformer-based review credibility and aspect sentiment extraction, and the multi-criteria evidence-fusion engine that produces the final Buy, Wait, or Avoid decision report.

4.8.1 FASTAPI REQUEST HANDLING

The FastAPI application acts as the REST communication boundary between the React/Next.js frontend and the Python AI decision engine. It validates the incoming product query, user budget, mandatory requirements, and priority weights, invokes the evaluation pipeline, and returns a structured JSON response.

Code snippet:

```python
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
from buysure_engine import evaluate_product_request

app = FastAPI(title="BUYSURE AI Decision API")

class ProductEvaluationRequest(BaseModel):
    product_query: str
    budget_max: float
    mandatory_needs: list[str] = []
    priority_weights: dict[str, float] = {}

@app.post("/api/evaluate")
def evaluate_endpoint(payload: ProductEvaluationRequest):
    result = evaluate_product_request(payload.dict())
    if "error" in result:
        raise HTTPException(status_code=404, detail=result["error"])
    return result
```

Explanation:

The FastAPI request-handling module defines a strongly typed *ProductEvaluationRequest* schema using Pydantic. When the frontend submits a POST request to */api/evaluate*, the endpoint validates that the budget and requirement fields are properly formatted, passes the payload to *evaluate_product_request()*, and returns the resulting evidence report or an HTTP 404 error if the product record cannot be resolved.

4.8.2 REVIEW CREDIBILITY AND ASPECT SENTIMENT ANALYSIS

```python
import numpy as np
import torch
import torch.nn.functional as F
from transformers import AutoTokenizer, AutoModelForSequenceClassification

# Load fine-tuned transformer for aspect sentiment & credibility scoring
tokenizer = AutoTokenizer.from_pretrained("buysure_review_model_v2")
model = AutoModelForSequenceClassification.from_pretrained("buysure_review_model_v2")
model.eval()

def compute_review_credibility(review_text, metadata):
    inputs = tokenizer(str(review_text), return_tensors="pt", truncation=True, max_length=256)
    with torch.no_grad():
        logits = model(**inputs).logits
    probs = F.softmax(logits, dim=1)
    text_cred = probs[0, 1].item()  # P(genuine | text)
    burst_penalty = 0.25 if metadata.get("is_burst_window") else 0.0
    dup_penalty = 0.35 * metadata.get("max_cosine_sim", 0.0)
    p_cred = max(0.05, min(1.0, text_cred - burst_penalty - dup_penalty))
    return round(p_cred, 4)

def aggregate_aspect_sentiment(reviews, aspect_keywords):
    aspect_scores = {}
    for aspect, kw_list in aspect_keywords.items():
        weighted_sum, weight_total = 0.0, 0.0
        for r in reviews:
            if any(k in r["text"].lower() for k in kw_list):
                w = r["credibility_weight"]
                weighted_sum += w * r["sentiment_polarity"]
                weight_total += w
        aspect_scores[aspect] = round(weighted_sum / max(weight_total, 1e-5), 3)
    return aspect_scores
```

The review intelligence module loads the fine-tuned sequence classification model and processes each customer review up to a maximum sequence length of 256 tokens. In *compute_review_credibility()*, the softmax probability of genuine text is combined with non-textual metadata penalties for submission burst windows (*is_burst_window*) and near-duplicate cosine similarity (*max_cosine_sim*) to produce a bounded credibility weight in *[0.05, 1.0]*.

In *aggregate_aspect_sentiment()*, reviews mentioning each product attribute (such as battery, camera, display, or thermals) are aggregated using their credibility weights *w*. This implements Equation (7) from the system design, ensuring that suspicious or repetitive bot comments have minimal influence on the product's aspect sentiment profile while genuine detailed reviews drive the feature scores.

4.8.3 EVIDENCE-FUSION DECISION AND ALTERNATIVE DISCOVERY

```python
def compute_buysure_decision(product, user_profile, evidence, catalog_df):
    # Step 1: Check hard mandatory requirements and strict budget constraint
    missing_needs = [req for req in user_profile["mandatory_needs"]
                     if not product["features"].get(req, False)]
    budget_ok = product["price"] <= user_profile["budget_max"]

    # Step 2: Compute weighted Evidence-Fusion Score (Equation 11)
    M_p = sum(w * product["norm_specs"].get(k, 0.0)
              for k, w in user_profile["priority_weights"].items())
    R_p = evidence["credibility_weighted_sentiment"]
    V_p = evidence["value_for_money_score"]
    Q_p = evidence["rating_consistency_score"]
    K_p = evidence["suspicion_risk_penalty"] + evidence["seller_risk"]
    raw_score = 0.30*M_p + 0.30*R_p + 0.20*V_p + 0.20*Q_p - 0.25*K_p
    buysure_score = round(max(0.0, min(100.0, raw_score * 100)), 2)

    # Step 3: Assign interpretable Buy / Wait / Avoid classification
    if missing_needs or K_p >= 0.45:
        verdict = "AVOID"
    elif not budget_ok or evidence.get("price_spike_pct", 0) > 5.0 or buysure_score < 75.0:
        verdict = "WAIT"
    else:
        verdict = "BUY"

    alternatives = find_better_alternatives(catalog_df, product, user_profile, top_n=5)
    return {
        "product": product["title"], "verdict": verdict,
        "buysure_score": buysure_score, "alternatives": alternatives
    }
```

The decision fusion module combines all normalized product and review signals to generate the final recommendation. First, *compute_buysure_decision()* checks whether the product satisfies every hard mandatory requirement in *user_profile["mandatory_needs"]* and whether its price lies within *budget_max*. Next, it computes the linear evidence-fusion score combining requirement match (*M_p*), credibility-weighted review sentiment (*R_p*), value-for-money (*V_p*), and rating consistency (*Q_p*), minus the composite risk penalty (*K_p*).

Finally, the rule layer assigns **AVOID** if any mandatory requirement fails or if risk exceeds the safety threshold, **WAIT** if the price is currently spiking or exceeds the strict budget, and **BUY** when all constraints pass with a strong overall score. The function also invokes *find_better_alternatives()* to return up to five competing products with higher requirement fit.

4.9 CHAPTER SUMMARY

This chapter presented the end-to-end implementation and experimental validation of the BUYSURE AI system. It detailed the collection and cleaning of the 18,400-review multi-category dataset, the multi-signal review credibility scoring pipeline, aspect-level sentiment aggregation, and the constraint-aware evidence-fusion decision equations. The quantitative evaluation across four experimental approaches demonstrated how integrating review credibility weights with personalized priority matching improved decision agreement from 61.0% to 86.4% and suspicious-review F1-score to 0.85. Finally, the live web application demonstration (https://buysure-ai.vercel.app) and core Python/FastAPI code modules illustrated how user queries are transformed into transparent Buy, Wait, and Avoid shopping reports.

CHAPTER 5. RESULTS AND DISCUSSION

5.1 INTRODUCTION

This chapter presents the experimental findings obtained from the BUYSURE AI decision-support system and discusses the performance of its review credibility, aspect extraction, and evidence-fusion modules. The system evaluates e-commerce products and classifies purchase recommendations into three primary user-facing verdicts—**Buy**, **Wait**, and **Avoid**—supported by six fine-grained operational decision sub-classes. Performance is analyzed using overall decision agreement accuracy, suspicious-review F1-scores, per-category F1 metrics, and the six-class confusion matrix, alongside qualitative findings from the live web application.

5.2 SYSTEM AND MODEL PERFORMANCE EVALUATION

The performance of BUYSURE AI was evaluated on a held-out benchmark of 3,680 customer reviews and 617 controlled multi-criteria buyer scenarios spanning six product categories. The final evidence-fusion model achieved an overall decision agreement accuracy of 86.4% on the three-class Buy/Wait/Avoid benchmark (and 84.0% exact match across all six fine-grained operational sub-classes), while achieving an F1-score of 0.85 on suspicious-review identification and 0.86 on aspect opinion extraction.

These results confirm that decomposing product evaluation into separate review-credibility, aspect-sentiment, and constraint-matching stages yields highly consistent purchase recommendations. The confusion matrix further demonstrates that hard constraint gating virtually eliminates high-risk false-positive Buy recommendations.

5.3 COMPARISON OF EXPERIMENTAL APPROACHES

Four progressive experimental configurations were evaluated to quantify the contribution of each architectural layer in BUYSURE AI. The experiments were designed to test whether adding NLP sentiment, multi-signal fake-review filtering, and personalized constraint fusion provides measurable value over standard marketplace star ratings.

In Experiment 1 (Rating-Only Baseline), products were ranked solely by their raw marketplace star rating and price. This baseline achieved only 61.0% decision agreement with human reference evaluations and obtained a 0.00 F1-score on suspicious-review detection, frequently recommending products with inflated promotional ratings or missing mandatory features. In Experiment 2 (Unweighted Text Sentiment), a transformer sentiment classifier analyzed review text without credibility weighting, improving decision agreement to 71.5% and suspicious-review F1 to 0.48.

In Experiment 3 (Multi-Signal Credibility ML), linguistic stylometrics were combined with temporal burst detection, cosine duplication screening, and rating-sentiment divergence. This configuration increased decision agreement to 81.2% and raised the suspicious-review F1-score to 0.79, effectively neutralizing repetitive bot bursts.

Finally, Experiment 4 (Full Evidence Fusion) integrated credibility-weighted aspect sentiment with hard user constraint satisfaction, budget-fit penalties, and priority weighting. This complete pipeline achieved 86.4% decision agreement accuracy and a 0.85 suspicious-review F1-score. The progression from Experiment 1 (61.0%) to Experiment 4 (86.4%) demonstrates that neither raw ratings nor isolated NLP sentiment is sufficient on its own; trustworthy shopping support requires combining review authenticity with explicit buyer priorities.

5.4 CONFUSION MATRIX AND CLASS-WISE RESULTS

The confusion matrix (Figure 4.4) was used to inspect the prediction behavior of the final BUYSURE AI engine across the 617 test scenarios. Along the diagonal, the system correctly identified 142 of 160 *Buy (Strong Fit)* cases (88.8%), 78 of 96 *Wait (Price/Timing)* cases (81.3%), 84 of 110 *Wait (Mixed/Trade-off)* cases (76.4%), 96 of 110 *Avoid (Spec Mismatch)* cases (87.3%), 68 of 78 *Avoid (High Risk/Fake)* cases (87.2%), and 50 of 63 *Alternative Recommended* cases (79.4%).

An examination of off-diagonal entries shows that most differences occur between adjacent caution states—such as a product with both a minor price markup and mixed secondary reviews being assigned *Wait (Price/Timing)* instead of *Wait (Mixed/Trade-off)*. Because both sub-classes map to a **Wait** recommendation in the user interface, the practical user-facing agreement reaches 86.4% (533 of 617 scenarios).

Most importantly, out of 188 total *Avoid* scenarios (covering hard specification failures, thermal throttling defects, and severe fake-review manipulation), only 3 scenarios were misclassified as *Buy*. This low critical-error rate (1.6%) highlights the effectiveness of evaluating mandatory pass/fail constraints and suspicion risk thresholds before computing weighted preference scores.

Overall, the class-wise evaluation confirms that BUYSURE AI is conservative when evidence is noisy or incomplete, preferring to issue a transparent **Wait** or **Avoid** warning rather than an overconfident **Buy** verdict.

5.5 WEB APPLICATION RESULTS

The deployed BUYSURE AI web application (https://buysure-ai.vercel.app) provides an interactive interface for personalized product evaluation, review credibility auditing, and side-by-side comparison. Users can submit a product query, specify their budget and priority features, and receive a structured decision report in under two seconds for catalog-indexed items.

The interface displays the predicted verdict badge (**BUY**, **WAIT**, or **AVOID**), the 0–100 BUYSURE confidence bar, budget compliance status, and five ranked evidence insights. When a product is flagged for thermal issues, inflated pricing, or suspicious review bursts, the application highlights the exact risk factor and provides direct comparison links to superior alternatives.

During demonstration testing across smartphones, laptops, and wireless audio gear (Figures 4.5 to 4.10), the web application accurately distinguished between high-value verified products (such as the Motorola Moto G67 5G and Lenovo IdeaPad Slim 5), temporarily overpriced listings (Samsung Galaxy S24 FE), and high-risk thermal or review-manipulated listings (POCO X7 5G and boAt Nirvana Ion ANC).

By translating complex NLP credibility weights and multi-criteria scoring equations into concise, human-readable evidence rows, the web application enables everyday shoppers to understand *why* a recommendation was made without inspecting raw model probabilities.

5.6 LIMITATIONS OF THE RESULTS

Although BUYSURE AI demonstrates strong accuracy in review credibility filtering and personalized purchase decision support, several limitations should be noted when interpreting the results.

First, the system achieves an overall decision agreement of 86.4%, meaning that roughly 13.6% of borderline trade-off scenarios still differ from human expert judgments—particularly when two competing products have nearly identical specifications and mixed subjective reviews.

Second, while the suspicious-review classifier achieves a 0.85 F1-score, highly sophisticated human-written promotional reviews that mimic balanced pros and cons without temporal burst patterns remain difficult to detect from text and basic metadata alone.

Third, the current evaluation relies on English-language reviews and structured specification tables within the project's 1,250-product benchmark. Products with zero existing reviews, unindexed niche categories, or heavily code-mixed regional reviews (such as Tanglish or Hinglish) may yield lower evidence confidence until multilingual and multimodal extensions are incorporated.

Finally, because e-commerce prices and seller inventories change continuously, a **Buy** or **Wait** recommendation is valid strictly for the price observed at analysis time. Integrating live automated price-drop webhooks across external marketplaces represents a key next step for production scaling.

5.7 CHAPTER SUMMARY

This chapter analyzed the quantitative and qualitative results of the BUYSURE AI system. Across 617 evaluation scenarios and 18,400 reviews, the full evidence-fusion pipeline achieved 86.4% decision agreement accuracy and a 0.85 suspicious-review F1-score, substantially outperforming raw star-rating and unweighted sentiment baselines. Confusion matrix analysis confirmed a 98.4% safety rate against recommending high-risk or incompatible products.

The live web application demonstration (https://buysure-ai.vercel.app) verified the complete user workflow across Buy, Wait, Avoid, Credibility Audit, Comparison, and Alternative Discovery scenarios. The chapter also documented current limitations regarding borderline subjective trade-offs, adversarial review phrasing, and dynamic marketplace pricing, providing a clear foundation for the conclusions and future enhancements presented in Chapter 6.

CHAPTER 6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

BUYSURE AI was developed as an intelligent, explainable shopping assistant for trustworthy product evaluation and personalized purchase decision support. The system addresses a pervasive challenge in modern online commerce: consumers face an overwhelming volume of product listings and reviews but lack reliable tools to filter out fake review bursts, verify marketing claims against real user experiences, and align product specifications with their personal budget and priorities.

By separating raw marketplace data from derived evidence, the proposed seven-layer architecture combines specification normalization, multi-signal suspicious-review detection, credibility-weighted aspect sentiment analysis, constraint-first product comparison, and transparent evidence fusion. Evaluated across 1,250 products, 18,400 customer reviews, and 617 controlled buyer scenarios, the final model achieved 86.4% decision agreement accuracy and a 0.85 F1-score in flagging suspicious reviews.

Integrated with a responsive React, Next.js, and FastAPI web application deployed at https://buysure-ai.vercel.app, BUYSURE AI delivers an immediately understandable 0–100 score and a clear **Buy**, **Wait**, or **Avoid** verdict backed by traceable pros, cons, risk alerts, and better-matching alternatives. While the system should be used as an evidence-based decision aid rather than an absolute guarantee of product performance, it demonstrates how combining NLP credibility auditing with personalized multi-criteria evaluation can make online shopping significantly safer, faster, and more transparent.

6.2 FUTURE ENHANCEMENTS

The BUYSURE AI platform is designed with a modular architecture that supports several high-impact future extensions:

1. **Multimodal Product Image and Screenshot Understanding:** Integrating vision-language and OCR models so users can upload mobile screenshots of any shopping app or product box to automatically extract specifications, hidden fine print, and pricing details.
2. **Real-Time Price Tracking and Target Alerts:** Adding a time-series price monitoring service that tracks historical price fluctuations across marketplaces and notifies users automatically when a **Wait**-listed product drops below their target buy price.
3. **Graph-Based Seller Syndicate and Risk Detection:** Expanding seller risk analysis using graph relational modeling (GraphRAG) to identify coordinated shell-seller networks, warranty fulfillment risks, and return-policy traps.

1. **Browser Extension for In-Page Overlay:** Packaging the BUYSURE AI API into a lightweight Chrome and mobile browser extension that injects the Buy/Wait/Avoid decision badge, credibility-adjusted rating, and alternative links directly onto live Amazon, Flipkart, and Croma product pages.
2. **Physics and Hardware Claim Verification:** Incorporating domain-specific engineering rules and benchmark lookup tables to cross-check exaggerated marketing numbers (such as physically implausible battery density or charger wattage claims) against verified OEM filings.
3. **Continuous User Feedback Calibration:** Allowing buyers to submit post-purchase satisfaction feedback with versioned privacy controls to continuously refine aspect priority weights and category-specific decision thresholds.

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